E10 on the new real-plant rows (3,000-row upload)

Maize and sorghum: zero-shot and real-row fine-tuning results

CropCode

2026-11-13

Headline

- **Maize** (130 held-out test plants, 13,500 camera stops, all three inputs): Chamfer distance (CD) 116.6 (headline model, zero-shot) $\rightarrow$ **23.6** after real-row fine-tuning (15 ep), $10^{-4}\,\mathrm{m}^2$. Median height error 52 % $\rightarrow$ **7.7 %**; leaf count within ± 1 on 22 % $\rightarrow$ **68 %** of stops; per-leaf length error 33 $\rightarrow$ **9.0 cm**, width 10.7 $\rightarrow$ **4.9 cm**, internode 8.4 $\rightarrow$ **3.7 cm** (true means 67, 18 and 13 cm).
- **Sorghum** (13 held-out test scans, zoomed window): CD 547 (headline, zero-shot), 391 (isolated control) $\rightarrow$ **48.8** after real-row fine-tuning with zoomed-crop augmentation, 40 epochs (58.2 at 15 epochs). On the full tall strip the 15-epoch fine-tune reaches 98.7 (from 1,936); the zoom is the better input.
- **Sorghum height is not learned** (error $\approx$72 % before and after): the scans carry no procedural parameters, so the sorghum fine-tune has no recipe or height loss, and the written stem length stays near the synthetic ceiling while the real plants are 2.2–5.3 m tall. Read sorghum by its clouds.

1 The data

The 2026-11-13 upload replaces the first E10 rows. Each species has 3,000 rows $\times$ 18 camera stops (54,000 stops): a photograph and a labelled point cloud (80k points per row, metres, camera frame) per stop, with the middle plant as the target.

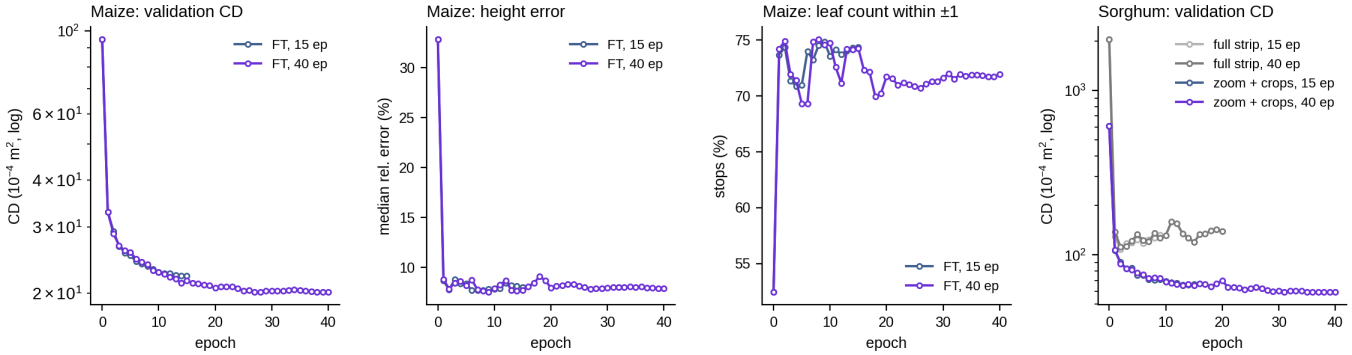
- **Maize**: 520 plant models; a per-leaf CSV for the middle plant (internode distance, leaf length, width, curvature, zenith, . . .). Split by plant: 351 train / 39 validation / 130 test.
- **Sorghum**: 52 scans, each with four tilted copies (208 plant models), heights 2.2–5.3 m; no leaf parameters. Split by scan (the tilted copies stay together): 34 / 5 / 13. The photographs are tall strips; the *zoomed* window is a square of side $\min(H, 3W)$ centred on the median image row of the middle plant's leaves, so the plant appears at maize-like scale. Training adds random zoomed crops (side $\times U(0.75, 1.25)$, centre between the 10th and 90th percentile of leaf rows, ± 10 % of the width sideways).



2 Fine-tuning runs

All runs start from the synthetic-row fine-tune (epoch 100) of their species and train on the training plants' stops (lr 2×10^{-5}, batch 16, 80 % of input tokens hidden); validation every epoch on the validation plants; the best validation checkpoint is tested.

Species	Run	Window	Epochs	Supervised	Val CD start	Best val CD
Maize	maize_real_ft_r3000	square	15 / 15	pc, depth, height, leafCount, stemInternodeSum, distance, leafLength, leafWidth, avgCurvature	94.7	22.3 (ep 15)
Maize	maize_real_ft_r3000_e40	square	40 / 40	pc, depth, height, leafCount, stemInternodeSum, distance, leafLength, leafWidth, avgCurvature	94.7	20.1 (ep 28)
Sorghum	sorghum_real_ft_r3000	full strip	9 / 15	pc, depth	2,040.1	108.4 (ep 2)
Sorghum	sorghum_real_ft_r3000_e40	full strip	20 / 40	pc, depth	2,040.1	111.8 (ep 2)
Sorghum	sorghum_real_ft_r3000_zoom	zoom + crops	15 / 15	pc, depth	605.9	65.7 (ep 14)
Sorghum	sorghum_real_ft_r3000_zoom_e40	zoom + crops	40 / 40	pc, depth	605.9	59.2 (ep 37)



3 Test results

Held-out test plants only. CD = squared Chamfer distance of the target plant's predicted cloud to its true cloud, $10^{-4}\,\mathrm{m}^2$ (lower is better; bold = best per input). Miss@5mm = share of true points with no prediction within 5 mm. Input: which streams the model sees (the others are hidden; the recipe is never an input). Per-leaf errors compare the written recipe with the leaf CSV, leaves matched by index from the base.

3.1 Maize, 130 test plants, 13,500 stops

Model	Input	CD [95 % CI] $10^{-4}\,\mathrm{m}^2$	Miss@5mm	Height err. median	Leaves ±1	Internode cm	Length cm	Width cm	Curv. 1/m
Headline model, zero-shot	all three	116.6 [111.4, 122.2]	0.957	52.3 %	22 %	8.4	33.0	10.7	2.29
	RGB	124.4 [119.5, 129.6]	0.948	47.0 %	35 %	7.6	32.8	10.7	2.11
	depth	149.8 [144.6, 155.2]	0.952	47.6 %	51 %	8.2	31.2	10.4	2.14
	cloud	123.3 [117.1, 129.7]	0.950	43.6 %	36 %	6.8	33.4	11.1	2.26
Isolated control, zero-shot	all three	119.2 [115.1, 123.9]	0.958	40.9 %	12 %	7.2	21.6	10.8	1.98
	RGB	204.5 [196.3, 213.4]	0.990	46.4 %	11 %	6.5	25.9	10.9	1.95
	depth	152.6 [147.8, 157.5]	0.960	41.6 %	27 %	6.6	24.0	10.6	1.95
	cloud	105.1 [99.5, 110.9]	0.962	39.3 %	47 %	6.7	31.5	11.1	2.09
Synthetic-row fine-tune, zero-shot	all three	95.8 [92.1, 100.1]	0.964	34.3 %	51 %	6.7	31.7	11.8	2.10
	RGB	117.2 [112.0, 123.0]	0.982	52.2 %	2 %	6.4	32.0	11.4	2.13
	depth	101.9 [98.7, 105.5]	0.963	46.8 %	8 %	6.3	29.3	11.5	2.00
	cloud	108.3 [102.3, 115.1]	0.977	38.5 %	26 %	6.5	31.1	11.5	2.09
Real-row fine-tune, 15 ep	all three	**23.6 [22.5, 24.8]**	0.914	7.7 %	68 %	3.7	9.0	4.9	1.84
	RGB	**40.6 [39.3, 42.1]**	0.956	28.3 %	46 %	4.3	12.0	5.3	1.90
	depth	**29.0 [27.7, 30.2]**	0.924	25.6 %	61 %	3.9	10.0	5.1	1.85
	cloud	**68.4 [64.9, 72.7]**	0.951	21.6 %	45 %	4.2	12.7	5.4	1.87

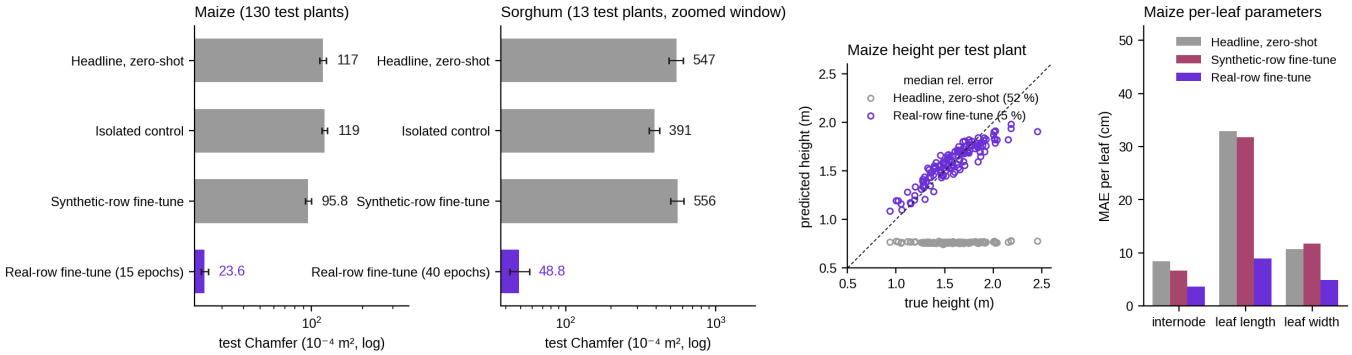
3.2 Sorghum, zoomed window, 13 test scans, 13,501 stops

Model	Input	CD mean [95 % CI] $10^{-4}\,\mathrm{m}^2$	CD median $10^{-4}\,\mathrm{m}^2$	Miss@5mm	Height err.* median
Headline model, zero-shot	all three	546.5 [487.7, 607.9]	457.1	0.980	71.0 %
	RGB	438.4 [410.8, 462.7]	328.6	0.979	62.7 %
	depth	375.5 [349.0, 402.9]	311.0	0.976	67.4 %
	cloud	703.1 [618.2, 795.2]	632.0	0.984	63.8 %
Isolated control, zero-shot	all three	390.9 [360.3, 421.8]	357.3	0.991	75.6 %
	RGB	582.6 [543.9, 620.6]	499.0	0.993	61.0 %
	depth	744.7 [692.0, 798.7]	643.4	0.996	61.6 %
	cloud	459.4 [408.2, 513.5]	408.6	0.978	62.8 %
Synthetic-row fine-tune, zero-shot	all three	556.1 [501.2, 613.1]	510.9	0.997	83.1 %
	RGB	729.6 [675.9, 785.4]	594.8	0.997	55.1 %
	depth	728.8 [678.1, 776.7]	607.3	0.998	57.6 %
	cloud	524.4 [464.7, 585.4]	484.0	0.986	71.6 %
Real-row fine-tune + crops, 15 ep	all three	58.2 [52.0, 65.8]	50.1	0.939	72.4 %
	RGB	139.0 [122.2, 161.3]	93.5	0.964	58.8 %
	depth	90.8 [78.7, 106.6]	50.9	0.912	55.2 %
	cloud	201.5 [184.0, 218.4]	174.7	0.976	69.5 %
Real-row fine-tune + crops, 40 ep	all three	**48.8 [42.1, 57.4]**	41.0	0.931	72.4 %
	RGB	**126.6 [111.1, 146.5]**	82.1	0.962	66.1 %
	depth	**80.9 [71.0, 92.3]**	43.6	0.906	63.2 %
	cloud	**182.0 [169.8, 194.5]**	156.2	0.975	68.3 %

3.3 Sorghum, full tall strip, 13 test scans

Model	Input	CD mean [95 % CI] 10^{-4} m^2	CD median 10^{-4} m^2	Miss@5mm	Height err.* median
Headline model, zero-shot	all three	1,936.2 [1,642.2, 2,235.4]	1,831.8	0.988	73.9 %
	RGB	844.2 [772.5, 918.1]	745.4	0.993	58.9 %
	depth	1,364.6 [1,195.6, 1,532.6]	1,276.2	0.994	72.1 %
	cloud	3,977.4 [3,297.8, 4,688.4]	3,897.9	0.988	64.3 %
Isolated control, zero-shot	all three	1,294.4 [1,013.3, 1,602.6]	1,180.4	0.991	79.4 %
	RGB	1,552.7 [1,410.4, 1,724.4]	1,463.9	0.998	53.6 %
	depth	2,353.3 [2,170.5, 2,562.4]	2,287.0	0.999	71.3 %
	cloud	3,163.6 [2,600.2, 3,757.3]	3,068.2	0.992	65.2 %
Synthetic-row fine-tune, zero-shot	all three	2,079.1 [1,701.4, 2,482.6]	2,033.2	0.998	85.2 %
	RGB	1,635.8 [1,511.1, 1,760.3]	1,550.6	0.999	33.8 %
	depth	1,495.6 [1,377.1, 1,621.7]	1,402.1	1.000	41.3 %
	cloud	3,161.1 [2,547.6, 3,812.0]	3,166.4	0.983	73.7 %
Real-row fine-tune, 15 ep	all three	**98.7 [92.8, 105.1]**	87.9	0.966	73.1 %
	RGB	**112.3 [104.8, 120.9]**	98.1	0.966	54.6 %
	depth	**105.2 [94.8, 118.6]**	90.8	0.961	53.2 %
	cloud	**560.7 [432.1, 710.7]**	490.9	0.983	61.2 %

*Sorghum height is read from the written stem length, which no sorghum run is trained on (no recipe truth); it stays near the synthetic ceiling against 2.2–5.3 m real plants. The zoomed and full-strip CDs are on different windows and are not comparable with each other.

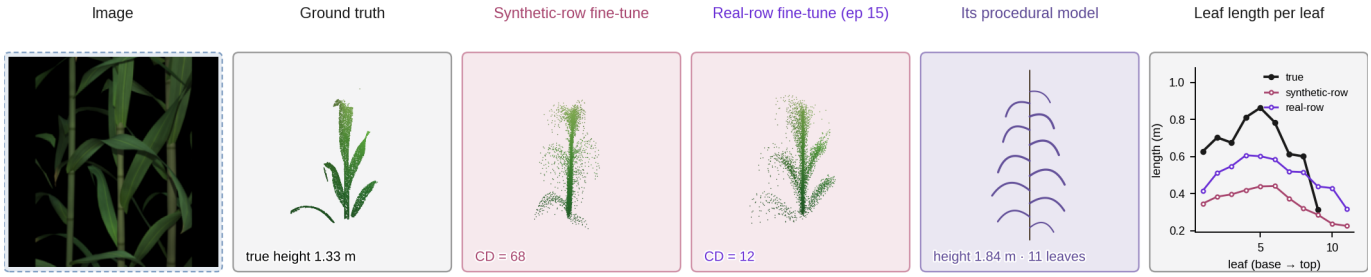


Left two: test CD of every model (all three inputs, 95 % CI). Third: maize height per test plant (median over its stops); the zero-shot headline model writes about 0.77 m for every plant. Right: maize per-leaf error.

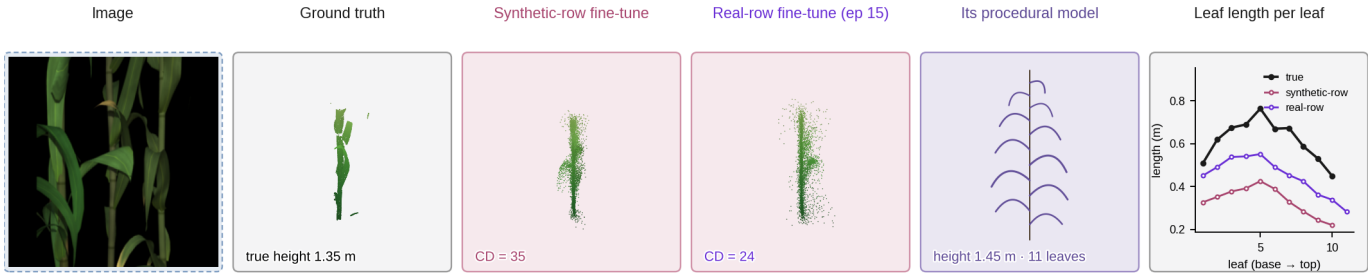
4 Examples

Test plants at the 25th, 50th and 75th percentile of the fine-tuned model's per-plant CD, each at its median-error stop. Columns: the photograph, the true plant, the synthetic-row fine-tune (before), the real-row fine-tune (after), the procedural model the real-row fine-tune writes, and (maize) its leaf lengths against the true ones.

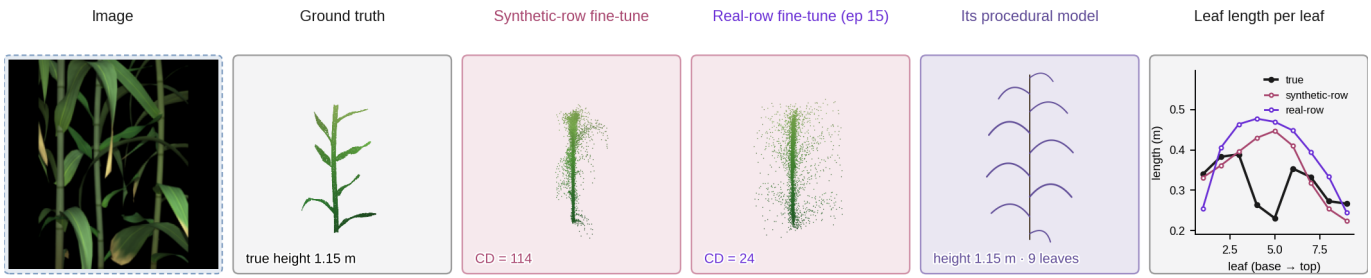
Maize (real-row fine-tune, 15 epochs)



25th percentile: test plant 0366, row 2649, stop 4; 11 % of the plant hidden by neighbours; true height 1.33 m, 9 leaves; written 1.84 m, 11 leaves



50th percentile: test plant 0359, row 1365, stop 18; 8 % of the plant hidden by neighbours; true height 1.35 m, 10 leaves; written 1.45 m, 11 leaves



75th percentile: test plant 0042, row 895, stop 16; 18 % of the plant hidden by neighbours; true height 1.15 m, 9 leaves; written 1.15 m, 9 leaves

Sorghum (zoomed window, real-row fine-tune + crops, 40 epochs)



25th percentile: test plant d2_452700, row 417, stop 7; 8 % of the plant hidden by neighbours; true height 4.12 m; written 1.10 m



50th percentile: test plant p1d1V_329706, row 879, stop 15; 11 % of the plant hidden by neighbours; true height 4.14 m; written 1.07 m



75th percentile: test plant d2_525769, row 1553, stop 1; 14 % of the plant hidden by neighbours; true height 4.23 m; written 1.04 m

5 Caveats and status

- The first E10 upload gave maize height error 1.6 % after fine-tuning; these rows give 7.7 % per stop (5 % per plant, median over its stops): a harder set (more plants, taller, more varied).
- Sorghum has 13 test scans (52 scans in total), so its confidence intervals are wide.
- The paper's Sec. 4.8 still reports the first upload.
- Not yet in this report (still running): job 17437008, job 17437009. Re-run figures/make_e10_r3000_report.py when they finish.